Merchant Api Document v1.0.9

General notes

- All requests must include an HTTP header with the key **X-Api-Key** and the appropriate API key. This key will be provided to you upon account creation
- HTTP response codes:
  - **2xx**: Indicates successful responses.
  - **4xx**: Indicates client-side errors (e.g., invalid request).
  - **5xx**: Indicates server-side errors.
- `{{baseUrl}}` is a placeholder for the environment-specific base URL, which will be provided to you when you are ready to integrate into a particular environment.
- Each API request includes a trace identifier, which can be found in the response headers under the key **X-Trace-Id**. If you need to contact us about a specific request, please include this trace identifier to help expedite the resolution process.
- For testing purposes, please reach out to obtain the latest customer app.
- You can use a deep link to redirect users to the InnBucks app as part of the user flow. The deep link URL format is as follows:

  `com.innbucks.customer://purchase?paymentToken={paymentToken}`

  In this example, `paymentToken=701564` represents the InnBucks code generated by the API call: `{{baseUrl}}/api/code/generate`
- `{{baseUrl}}` is a placeholder for the specific environment base url that will be shared with you once you are ready to integrate into a specific environment

Authentication

General notes

- The system uses **Bearer Authentication** for all secured endpoints
- To access secured endpoints, include a **Bearer token** in the **Authorization** header of your request

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- The header should be formatted as follows:

  Authorization: Bearer {{accessToken}}

  Replace ***{{accessToken}}*** with the token received in the login response.

Response Code

- **00** is a success response code
- For any other response code, the request has failed. The specific reason for the failure can be found in the **responseDescription** field

1. Login

This API is used to authenticate with the system and obtain an access token. By default, the access token expires after 15 minutes (unless otherwise specified for a specific environment). The token must be included as a bearer token in the authorization header for all secure API calls.

Request

- **Method**: POST
- **Endpoint**: ***{{baseUrl}}/auth/third-party***

Sample Request

```
{
  "username": "testClient45HUj",
  "password": "password"
}
```

Sample Response

```
{
  "accessToken": "accessTokenValue",
  "responseCode": "00",
  "responseDescription": "Auth successful"
}
```

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Requests

General notes

- All requests listed below are secured and require a valid bearer token.
- If a reference field is supported for a request, it is intended to be used as a reference from the source system of the request. This field is optional; if not provided, the system will generate it automatically. The reference will also appear as the **stan** in the response.
- All amount-related fields are in cents, as currency is represented in minor denominations.
- **Response code 0** indicates a successful response unless specifically specified for a request . Any other response code indicates a failure, and the reason will be detailed in the **responseMsg** field.
- A complete list of response codes is available at: `{{baseUrl}}/api/file/response-codes` (**GET** request).

1. InnBucks Code Generation

This API allows a merchant to generate an InnBucks code for a customer to use for payments or withdrawals. The merchant should display the QR code (provided as a base64 string in the **qrCode** field of the response) along with the plain InnBucks code to enable the customer to initiate the payment via USSD or the mobile app.

Request

- **Method**: POST
- **Endpoint**: `{{baseUrl}}/api/code/generate`

Parameters

- **type**: The type of InnBucks code to generate and this can be either **PAYMENT** or **WITHDRAWAL**.
- The type of code you can generate depends on the merchant type and the allowed transactions.

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Sample Request

```
{
  "reference": "1234",
  "narration": "string",
  "amount": 1,
  "type": "PAYMENT"
}
```

Sample Response

```
{
  "stan": "414107",
  "authNumber": "1616800",
  "processedDateTime": "2020-07-09 13:30:53.515",
  "responseCode": 0,
  "responseMsg": "Approved or completed successfully",
  "code": "701285660",
  "qrCode": "base64 string",
  "amount": "100",
  "description": "string"
}
```

2. InnBucks Code Inquiry

This API retrieves the status of a generated InnBucks code associated with a merchant. If request was successful, the **status** field in the response indicates the current state of the transaction, which can be one of the following:

- **New**: Code is generated but not yet claimed or paid.
- **Claimed**: Code has been finalised by the customer.
- **Paid**: Code has been finalised by the customer.
- **Expired**: Code validity period elapsed.
- **Timed Out**: Transaction window exceeded.

Request

- **Method**: POST
- **Endpoint**: `{{baseUrl}}/api/code/query`

Sample Request

```
{
  "reference": "1234",
```

```
  "code": "61265602"
}
```

Sample Response

```
{
  "stan": "1655116056",
  "authNumber": "7875",
  "processedDateTime": "2022-07-05 12:11:47.067",
  "responseCode": 0,
  "responseMsg": "Approved or completed successfully",
  "code": "701848897",
  "amount": "100",
  "status": "New",
  "timeToLive": "576sec",
  "description": "string"
}
```

3. InnBucks Code Inquiry By Original Reference

This API retrieves the status of a generated InnBucks code associated with a merchant. If request was successful, the **status** field in the response indicates the current state of the transaction, which can be one of the following:

- **New**: Code is generated but not yet claimed or paid.
- **Claimed**: Code has been finalised by the customer.
- **Paid**: Code has been finalised by the customer.
- **Expired**: Code validity period elapsed.
- **Timed Out**: Transaction window exceeded.

Parameters

- **originalReference**: this is the authNumber received in the code generation response

Request

- **Method**: POST
- **Endpoint**: `{{baseUrl}}/api/code/query/originalReference`

Sample Request

```
{
```

```
  "reference": "890790",
  "originalReference": "1616800"
}
```


Sample Response


```
{
  "stan": "1655116056",
  "authNumber": "7875",
  "processedDateTime": "2022-07-05 12:11:47.067",
  "responseCode": 0,
  "responseMsg": "Approved or completed successfully",
  "code": "701848897",
  "amount": "100",
  "status": "Claimed",
  "timeToLive": "0sec",
  "description": "Cash Send, Testing"
}
```


4. Merchant Mini Statement

This API retrieves recent InnBucks code transactions for a merchant.

Request

- **Method**: GET
- **Endpoint**: ***{{baseUrl}}/api/code/{{accountId}}/miniStatement***
- **Note**: Replace the ***{{accountId}}*** placeholder with your specific account number.

Sample Response


```
{
  "stan": "1655116056",
  "authNumber": "7880",
  "processedDateTime": "2022-07-05 12:16:32.794",
  "responseCode": 0,
  "responseMsg": "Approved or completed successfully",
  "code": [
    {
      "amount": "100",
      "code": "701848897",
      "codeType": null,
      "createDate": "2022-07-05 12:11:23",
      "closedDate": null,
```

```
      "createStan": "7873",
      "createStanley": "1847",
      "state": "Pending"
    },
    {
      "amount": "100",
      "code": "701977985",
      "codeType": null,
      "createDate": "2022-07-05 12:07:30",
      "closedDate": "2022-07-05 12:08:08",
      "createStan": "7866",
      "createStanley": "1838",
      "state": "Claimed"
    }
  ]
}
```


5. Account Statement

This API retrieves a full statement for a specific account within a selected period. The period is limited to the current banking rules, which allow only a calendar month

Request

- **Method**: POST
- **Endpoint**: `{{baseUrl}}/api/account/fullStatement`

Parameters

- ***startDateTime*** *and* ***endDateTime****: These fields should be provided in the following format:* `yyyy-MM-dd HH:mm:ss`

Sample Request

```
{
  "accountNumber": "1234",
  "startDateTime": "2022-06-01 00:00:00",
  "endDateTime": "2022-06-30 23:59:59",
}
```

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Sample Response


```
{
  "startDate": "2024-01-01T00:00:00",
  "endDate": "2024-01-31T23:59:59",
  "responseCode": 0,
  "responseMsg": "Approved or completed successfully",
  "reference": "1290786790",
  "accountDetails": {
    "accountNumber": "1234",
    "firstName": "Client Name",
    "lastName": "Client Surname",
    "mobileNumber": "263773123123",
    "currency": "USD"
  }
  "transactions": [
    {
      "type": "Opening Balance",
      "narration": "Opening Balance",
      "creditAmount": 100000,
      "debitAmount": null,
      "balance": 100000,
      "reference": "222556",
      "postingDate": "2024-01-01T00:00:00",
      "transactionDate": "2024-01-01T00:00:00",
       "additionalData": {
         "KEY": "VALUE"
       }
    },
   {
      "type": "Credit",
      "narration": "Transfer",
      "creditAmount": 27000,
      "debitAmount": null,
      "balance": 127000,
      "reference": "6789123",
      "postingDate": "2024-01-12T15:21:00",
      "transactionDate": "2024-01-12T15:21:00",
       "additionalData": {
         "KEY": "VALUE"
       }
    },
    {
      "type": "Debit",
      "narration": "Payment",
      "creditAmount": null,
```


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```
      "debitAmount": 2000,
      "balance": 125000,
      "reference": "6789123",
      "postingDate": "2024-01-13T17:45:00",
      "transactionDate": "2024-01-13T17:45:00",
       "additionalData": {
          "KEY": "VALUE"
        }
    },
   {
      "type": "Closing Balance",
      "narration": "Closing Balance",
      "creditAmount": 125000,
      "debitAmount": null,
      "balance": 125000,
      "reference": "785634",
      "postingDate": "2024-01-31T23:59:59",
      "transactionDate": "2024-01-31T23:59:59",
       "additionalData": {
          "KEY": "VALUE"
        }
    }
  ]
}
```


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6. Bank Change

This API is used when a cash transaction occurs, but the change cannot be settled immediately. If the customer requests, the change can be credited to their InnBucks account. This is subject to limits and can be used for amounts up to **US$5** (subject to change based on business conditions).

Request

- **Method**: POST
- **Endpoint**: ***{{baseUrl}}/api/transaction/bankChange***

Sample Request


```
{
  "reference": "1234",
  "amount": 1,
  "narration": "test transfer",
```

```
  "destinationMsisdn": "263770123123"
}
```

Sample Response

```
{
  "stan": "420032",
  "authNumber": "1735095",
  "processedDateTime": "2021-03-12 15:23:45.387",
  "responseCode": 0,
  "responseMsg": "Approved or completed successfully",
  "accountNumber": "123456789"
}
```

7. Linked Account Inquiry

This API retrieves the account details linked to a specific MSISDN.

Request

- **Method**: GET
- **Endpoint**: ***{{baseUrl}}/api/account/msisdn/{{msisdn}}/details***
- **Note**: The MSISDN should be provided in the format: **263772123123**
- **Note**: Replace the ***{{msisdn}}*** placeholder with the customer's msisdn.

Sample Response

```
{
  "firstName": "James",
  "lastName": "Mufambanaayo",
  "accountNumber": "123456789"
}
```

8. Account Deposit

This API is used to deposit funds into a customer's account. The deposit flow involves the following steps:

1. Retrieve the account details linked to the customer's MSISDN using the **Linked Account Inquiry** endpoint.

2. Manually verify the account details against the national ID provided by the customer for compliance purposes
3. Process the deposit.

Request

- **Method**: POST
- **Endpoint**: `{{baseUrl}}/api/transaction/deposit`

Parameters

- **destinationAccount**: The account number returned in the linked account inquiry response for the given MSISDN.
- **type** (Optional): A field to specify the type of deposit. This field is required if there are custom configurations needed for the deposit (e.g., special processing rules or limits).
- **reference: It can be used for duplicate checks only if duplicate check feature is**

Sample Request

```
{
  "reference": "FBNWIO2",
  "amount": 1,
  "narration": "test deposit",
  "destinationAccount": "123456789"
}
```

Sample Response

```
{
   "stan": "abcd-efghi-8979",
   "authNumber": "0123b025-2a5a-4fd8-8337-84133ea77d40.1763634254",
   "processedDateTime": "2021-03-12T12:39:19.641+0200",
   "responseCode": 0,
   "responseMsg": "Approved or completed successfully",
   "available": "2",
   "code": null,
   "currency": "USD",
   "debitAccountNumber": "2000473339827",
```

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```
    "creditAccountNumber": "123456789",
    "reference": "test depo",
    "ledger": "2",
    "amount": "2",
    "currencySymbol": "$ (values are in subunit)"

}
```

9. Utility Payment

This API processes a utility payment for a specified provider and product.

Request

- **Method**: POST
- **Endpoint**: `{{baseUrl}}/api/utility/provider/payment`

Parameters

- **narration** (Optional): A field for any additional description or notes related to the payment.
- **additionalData** (Optional): A field for any extra data required by the provider or specific product

Sample Request

```
{
  "provider": "ECONET",
  "providerProduct": "AIRTIME",
  "amount": 100,
  "currency": "USD",
  "reference": "123456789abc",
  "narration": "utility payment",
  "destinationAccount": "123456789",
  "additionalData": {
    "KEY": "VALUE"
  }
}
```

Sample Response

```
{
  "stan": "591844",
  "authNumber": "249878909",
  "processedDateTime": "2021-09-05 10:48:08.040",
```

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```
  "responseCode": 0,
  "responseMsg": "Approved or completed successfully",
  "fees": [],
  "amount": "100",
  "ledger": "2500",
  "currency": "USD",
  "available": "2500",
  "dynamicData": {
    "reference": "56677493",
    "instructions": "Transaction Amount USD1.00",
    "notifications": {}
  },
  "commissions": [
    {
      "name": "Commission",
      "description": "DebitCommission",
      "amount": "5"
    }
  ],
  "accountNumber": "123456789",
  "currencySymbol": "$ (values are in subunit)"
}
```


10. Reversal

This feature is only available for deposit transactions. Real-time reversals are not available for any code-based transactions. **Response code 00** (field responseCode in the response) indicates a successful request. Any other response code indicates a failure, and the reason will be detailed in the **responseDescription** field. Possible response code are below:

- 00 - Approved or completed successfully
- 025 - Unable to locate original request or already processed
- 096 - Request failed, please try again later

Request

- **Method**: POST
- **Endpoint**: `{{baseUrl}}/api/transaction/reversal/v2`

Parameters

- **amount**: Must match the amount from the original transaction request.

- **originalParticipantReference**: The reference of the original transaction.
- **participantReference**: A unique reference for the reversal.
- **type**: Should be set to "CREDIT".
- **destinationAccount**: The destination account used in the original transaction.

Sample Request

```
{
  "amount": 1,
  "currency": "USD",
  "type": "CREDIT",
  "destinationAccount": "123456789",
  "participantReference": "rev-FBNWIO2",
  "originalParticipantReference": "FBNWIO2"
}
```

Sample Response

```
{
  "reference": "588586383",
  "participantReference": "rev-FBNWIO2",
  "responseCode": "00",
  "responseDescription": "Approved or completed successfully"
}
```

11. Deposit Inquiry

This feature is only available for deposit transactions. This API checks the status of previous **deposit** transactions.

Request

- **Method**: POST
- **Endpoint**: `{{baseUrl}}/bank/api/transaction/inquiry`

Parameters

- ***accountNumber:*** Must match the destinationAccount used in the original transaction request
- ***originalParticipantReference****:* Corresponds to the **reference** provided in the original transaction request.
- ***participantReference*** : Is a unique reference for the transaction enquiry request.

Sample Request

```
{
  "accountNumber": "123456789",
  "participantReference": "inq-FBNWIO2",
  "originalParticipantReference": "FBNWIO2"
}
```

Sample Response

- if found, the original transaction is represented by the field **details** and it has combined fields of the related request and response

```
{
  "responseCode": "00",
  "responseDescription": "Approved or completed successfully",
  "reference": "678054",
  "participantReference": "inq-FBNWIO2",
  "details": {
    "responseCode": "00",
    "responseDescription": "Approved or completed successfully",
    "reference": "9001746",
    "participantReference": "FBNWIO2",
    “additionalData":{},
    "amount": 100,
    “dateTime":”2022-12-16 11:39:01.387000”
    "narration": "inbound remit test",
```

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"accountNumber": "3008608473674"
}
}

==========THE END==========

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